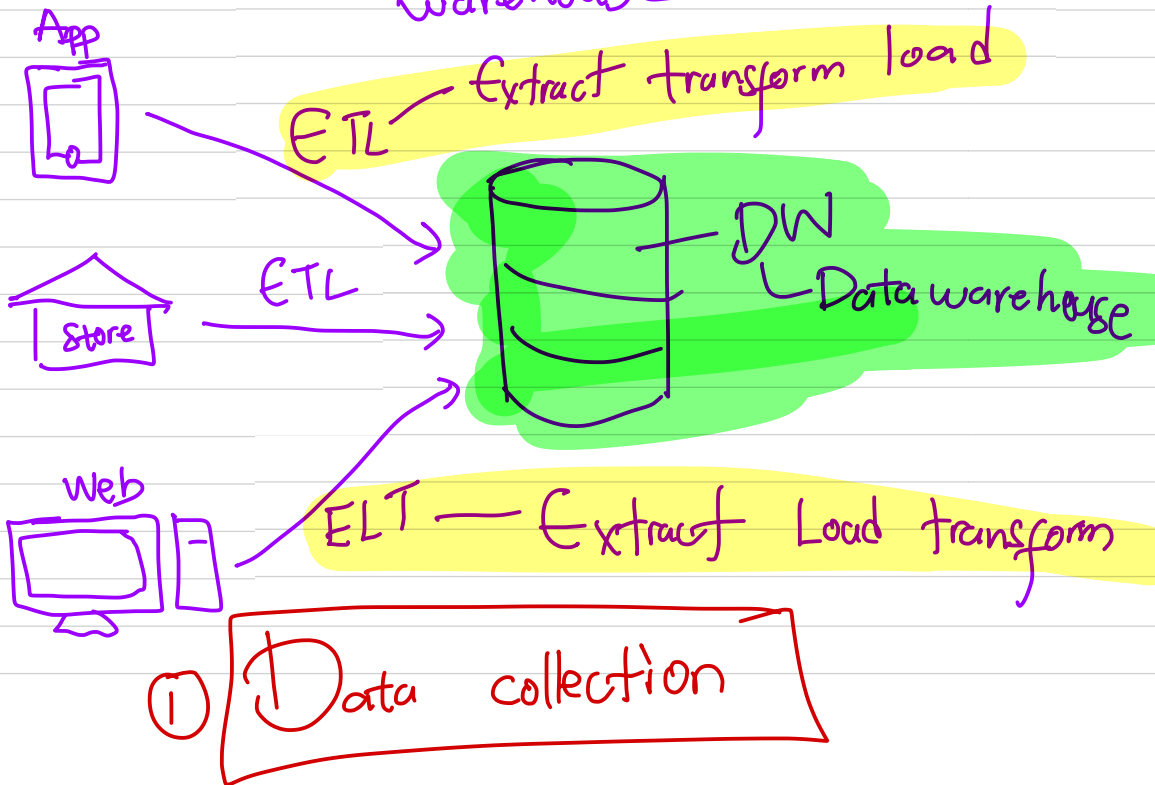


Machine Learning Process

- The outcome of a machine learning process is an AI tool/systems
- In order to build a machine learning model - we need data

The role of the Data Engineer is to make the data available in the datawarehouse

Build data pipelines to store the data in the warehouse



② Stakeholder: Define the problem

Before building AI/ML solution we need to define the problem we are trying to solve

Bank:

└ Predict the likelihood of a customer leaving my bank and going to another

Goal - we want to retain our customers.

FNB ~~→~~ Capitec

I want to create a machine learning model to predict the customers likely to churn

③ Explore and Understand data

└ Exploratory Data Analysis (EDA)



Correlation

You analyze the data to understand patterns, trends, and relationships.

Asking data driven Questions

- which customers are leaving?
- does age affect churn?
- Do customers with many complaints leave more

④ Prepare the Data for Machine Learning

Feature Build

- Converting text into numbers (Encoding)
- Scaling numeric values
- Creating new useful columns
- Splitting data into feature & target

Feature Engineering

feature Selection

⑤ Split the Data (Train-Test split)

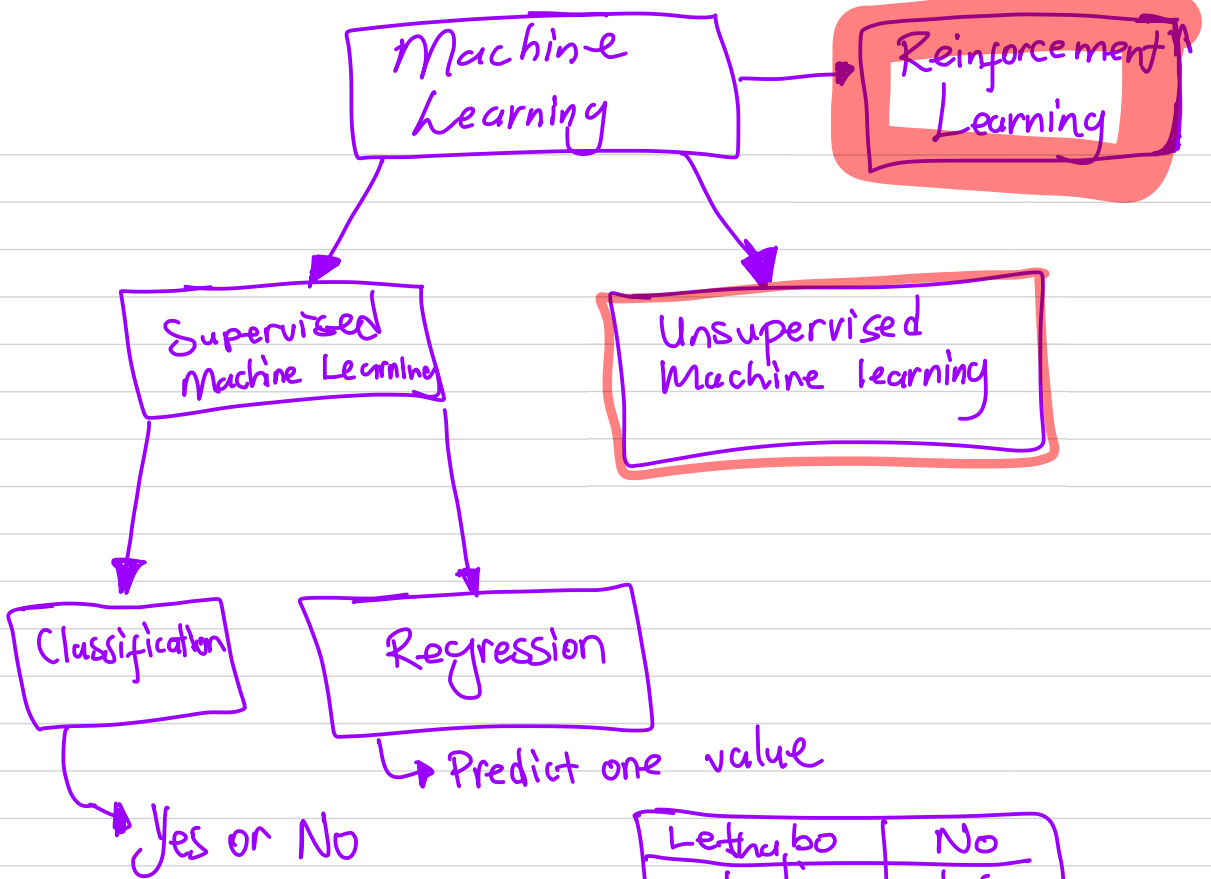
Training Data — used to teach the model (80%)

Testing Data — used to check if the model works (20%)

1	5
2	6
3	7
4	8

9
10

⑥ Choose Machine Learning Algorithm



Letnabo	No
mabali	Yes
Turni	Yes
Thuto	No